English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

English transcription of the handwritten solution notes. Original calculations are retained; apparent inconsistencies are identified in the transcription notes.

Problem 1

We have $\mathbb{E}X_i = 5011$ and $D(X_i) = 4000$.

1. The sum

Let $S_n = \sum_{i=1}^{n} X_i$. We seek n such that $\mathbb{P}(S_n > 10^6) \geq 99\%$. By the central limit theorem,

$$\begin{aligned}\mathbb{P}(S_n > 10^6) &= \mathbb{P}\left(\frac{S_n - \mathbb{E}S_n}{\sqrt{D(S_n)}} > \frac{10^6 - 5011n}{\sqrt{4000n}}\right) \\ &\approx \mathbb{P}\left(N(0,1) > \frac{10^6 - 5011n}{\sqrt{4000n}}\right).\end{aligned}$$

For this to be at least 99%, require

$$\frac{10^6 - 5011n}{\sqrt{4000n}} < -2.33, \qquad \text{giving} \qquad n \geq 200.$$

2. The maximum

We seek n such that

$$\mathbb{P}\left(\max_{1\leq i\leq n} X_i > 10^4\right) \geq 99\%.$$

Equivalently,

$$\mathbb{P}\left(\max_{1\leq i\leq n} X_i \leq 10^4\right) \leq 1\%, \qquad [\mathbb{P}(X_1 \leq 10^4)]^n \leq 1\%.$$

Writing $p = \mathbb{P}(X_1 \leq 10^4)$, this gives

$$n \geq \frac{\ln(1/100)}{\ln p}.$$

To obtain a particular value of p, a specific distribution of X_i must be chosen. If $X_i \sim N(5011, 4000)$, then

$$\mathbb{P}(X_1 \leq 10^4) = \mathbb{P}\left(N(0,1) \leq \frac{10^4 - 5011}{\sqrt{4000}}\right) \approx \mathbb{P}(N(0,1) \leq 78.88).$$

Thus the probability for the maximum involves $\Phi(78.88)^n$. All solutions that reach an expression of this kind receive the maximum possible number of points.

Bonus: an explicit bound

To find an explicit value of n such that $\Phi(78.88)^n \leq 1\%$, note that although $\Phi(78.88)$ is very close to 1, raising it to a sufficiently high power makes it close to 0. The standard bounds for $\mathbb{P}(N(0,1) > x) = 1 - \Phi(x)$ are

$$\left(\frac{1}{x} - \frac{1}{x^3}\right)\frac{e^{-x^2/2}}{\sqrt{2\pi}} \leq 1 - \Phi(x) \leq \frac{1}{x}\frac{e^{-x^2/2}}{\sqrt{2\pi}}.$$

Proof. Integrate by parts twice:

$$\begin{aligned}
1-\Phi(x) &= \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\,\mathrm{d}t \\
&= \frac{1}{\sqrt{2\pi}}\left(-\int_x^\infty \frac{1}{t}\,\mathrm{d}(e^{-t^2/2})\right) \\
&= \frac{1}{\sqrt{2\pi}}\left(\frac{1}{x}e^{-x^2/2} + \int_x^\infty e^{-t^2/2}\,\mathrm{d}(1/t)\right) \\
&= \frac{1}{\sqrt{2\pi}}\left(\frac{1}{x}e^{-x^2/2} - \int_x^\infty \frac{1}{t^2}e^{-t^2/2}\,\mathrm{d}t\right) \\
&= \frac{1}{\sqrt{2\pi}}\left(\frac{1}{x}e^{-x^2/2} + \int_x^\infty \frac{1}{t^3}\,\mathrm{d}(e^{-t^2/2})\right) \\
&= \frac{1}{\sqrt{2\pi}}\left(\frac{1}{x}e^{-x^2/2} - \frac{1}{x^3}e^{-x^2/2} - \int_x^\infty e^{-t^2/2}\,\mathrm{d}(1/t^3)\right).
\end{aligned}$$

The final line in the original is written

$$= \frac{1}{\sqrt{2\pi}}\left(\frac{1}{x}-\frac{1}{x^3}\right)e^{-x^2/2} - \frac{3}{\sqrt{2\pi}}\int_x^\infty \frac{1}{t^4}e^{-t^2/2}\,\mathrm{d}t \geq \frac{1}{\sqrt{2\pi}}\left(\frac{1}{x}-\frac{1}{x^3}\right)e^{-x^2/2}. \qquad \text{(Note 1)}$$

Hence, for $x = 78.88$,

$$\Phi(78.88) \leq 1 - \left(\frac{1}{x}-\frac{1}{x^3}\right)\frac{e^{-x^2/2}}{\sqrt{2\pi}}.$$

It is sufficient to find n such that

$$\left[1 - \left(\frac{1}{x}-\frac{1}{x^3}\right)\frac{e^{-x^2/2}}{\sqrt{2\pi}}\right]^n \leq 1\%.$$

The source next writes

$$n \ln\left[1 - \left(\frac{1}{x}-\frac{1}{x^3}\right)\frac{e^{-x^2/2}}{\sqrt{2\pi}}\right] \leq 1\%. \qquad \text{(Note 2)}$$

Using $\ln(1+u) \leq u$ (or $\ln(1-u) \sim -u$), the subsequent line is

$$n\left[-\left(\frac{1}{x}-\frac{1}{x^3}\right)\frac{e^{-x^2/2}}{\sqrt{2\pi}}\right] \leq \ln(1/100).$$

This gives the sufficient bound

$$n \geq \ln(100)e^{x^2/2}\sqrt{2\pi}\,\frac{x^3}{x^2-1}, \qquad \text{and thus} \qquad n \geq 1.16 \cdot 10^{1354}!$$

Problem 2

See Problem 3 of the Software Engineering examination, January 2020/2021.

Problem 3

Let $X \sim \mathrm{Exp}(\alpha^{-1})$ and $Y \sim \mathrm{Exp}(\beta^{-1})$.

1. Estimating the mean

We seek c such that $\mathbb{P}(|\alpha - \overline{X}| > c) \leq 1\%$. The handwritten source gives the following CLT normalization for large n:

$$\frac{\overline{X}-\alpha}{\alpha\sqrt{n}} \approx Z \sim N(0,1). \qquad \text{(Note 3)}$$

It therefore seeks c such that

$$\mathbb{P}\left(|Z| > \frac{c}{\alpha\sqrt{n}}\right) \le 1\% \quad \iff \quad \mathbb{P}\left(Z > \frac{c}{\alpha\sqrt{n}}\right) \le 0.5\%.$$

The source concludes

$$\frac{c}{\alpha\sqrt{n}} \approx 2.57, \qquad c \approx 2.57\alpha\sqrt{n}. \tag{Note 3}$$

2. The ratio

Set $Z_i = \sqrt{X_i/Y_i}$. If $\mathbb{E}Z_i < \infty$, the strong law of large numbers gives

$$\frac{Z_1 + \cdots + Z_n}{n} \longrightarrow \mathbb{E}Z_1 \qquad \text{almost surely.}$$

First method. For $t > 0$, using $\mathbb{P}(\text{Exp}(\lambda) > t) = e^{-\lambda t}$,

$$\begin{aligned}
\mathbb{P}\left(\sqrt{X/Y} > t\right) &= \mathbb{P}(X > Yt^2) = \int_0^\infty \mathbb{P}(X > yt^2) f_Y(y)\,\mathrm{d}y \\
&= \int_0^\infty e^{-yt^2/\alpha}\frac{1}{\beta}e^{-y/\beta}\,\mathrm{d}y \\
&= \left(\frac{t^2}{\alpha} + \frac{1}{\beta}\right)^{-1}\frac{1}{\beta}\int_0^\infty e^{-y(t^2/\alpha+1/\beta)}\,\mathrm{d}\left[y(t^2/\alpha + 1/\beta)\right] \\
&= \frac{\alpha\beta}{\beta t^2 + \alpha}\frac{1}{\beta} = \frac{\alpha}{\beta t^2 + \alpha}.
\end{aligned}$$

The integral after substitution equals $\int_0^\infty e^{-x}\,\mathrm{d}x = 1$. Therefore,

$$\begin{aligned}
\mathbb{E}\sqrt{X/Y} &= \int_0^\infty \mathbb{P}\left(\sqrt{X/Y} > t\right)\,\mathrm{d}t \\
&= \int_0^\infty \frac{\alpha}{\beta t^2 + \alpha}\,\mathrm{d}t = \int_0^\infty \frac{1}{(\beta/\alpha)t^2 + 1}\,\mathrm{d}t \\
&= \sqrt{\frac{\alpha}{\beta}}\int_0^\infty \frac{\mathrm{d}y}{1 + y^2} = \sqrt{\frac{\alpha}{\beta}}\frac{\pi}{2},
\end{aligned}$$

where $y = \sqrt{\beta/\alpha}\,t$. This is different from $\sqrt{\mathbb{E}X/\mathbb{E}Y} = \sqrt{\alpha/\beta}$.

Second method. Independence gives

$$f_{X,Y}(x,y) = f_X(x)f_Y(y) = \frac{1}{\alpha}e^{-x/\alpha}\frac{1}{\beta}e^{-y/\beta}.$$

Thus

$$\begin{aligned}
\mathbb{E}\sqrt{X/Y} &= \int_0^\infty\int_0^\infty \sqrt{x/y}\,\frac{1}{\alpha}e^{-x/\alpha}\frac{1}{\beta}e^{-y/\beta}\,\mathrm{d}x\,\mathrm{d}y \\
&= \left(\int_0^\infty \sqrt{x}\frac{1}{\alpha}e^{-x/\alpha}\,\mathrm{d}x\right)\left(\int_0^\infty \frac{1}{\sqrt{y}}\frac{1}{\beta}e^{-y/\beta}\,\mathrm{d}y\right).
\end{aligned}$$

Substitute $x' = x/\alpha$ and $y' = y/\beta$. The factors $\sqrt{\alpha}$ and $1/\sqrt{\beta}$ can be taken outside, giving

$$\mathbb{E}\sqrt{X/Y} = \sqrt{\frac{\alpha}{\beta}}\Gamma(3/2)\Gamma(1/2) = \sqrt{\frac{\alpha}{\beta}}\frac{1}{2}\Gamma(1/2)^2 = \sqrt{\frac{\alpha}{\beta}}\frac{\pi}{2}.$$

Here $\Gamma(z) = \int_0^\infty x^{z-1}e^{-x}\,\mathrm{d}x$ and $\Gamma(z+1) = z\Gamma(z)$.

Third method. Introduce $Z = \sqrt{X/Y}$ and $W = \sqrt{Y}$, so $X = Z^2W^2$ and $Y = W^2$. The Jacobian is

$$|J| = \begin{vmatrix} 2zw^2 & 2z^2w \\ 0 & 2w \end{vmatrix} = 4zw^3.$$

Thus, for $z, w > 0$,

$$f_{Z,W}(z,w) = f_{X,Y}(z^2w^2, w^2)\, 4zw^3 = \frac{1}{\alpha\beta} e^{-z^2w^2/\alpha - w^2/\beta}\, 4zw^3.$$

Integrating out w,

$$\begin{aligned} f_Z(z) &= \int_0^\infty \frac{1}{\alpha\beta} e^{-w^2(1/\beta + z^2/\alpha)}\, 4zw^3 \,\mathrm{d}w \\ &= \frac{4z}{\alpha\beta}\frac{1}{2}\int_0^\infty e^{-w^2(1/\beta+z^2/\alpha)} w^2 \,\mathrm{d}(w^2) \\ &= \frac{2z}{\alpha\beta}\left(\frac{1}{\beta} + \frac{z^2}{\alpha}\right)^{-2} \int_0^\infty e^{-y} y \,\mathrm{d}y \\ &= \frac{2z}{\alpha\beta}\frac{(\alpha\beta)^2}{(\alpha+\beta z^2)^2} = \frac{2\alpha\beta z}{(\alpha+\beta z^2)^2} = 2\frac{\beta}{\alpha}\frac{z}{(1+(\beta/\alpha)z^2)^2}. \end{aligned}$$

We used $y = w^2(1/\beta + z^2/\alpha)$ and $\int_0^\infty y e^{-y}\,\mathrm{d}y = \mathbb{E}[\mathrm{Exp}(1)] = 1$. Finally, with $z = \sqrt{\alpha/\beta}\, y$,

$$\begin{aligned} \mathbb{E}Z &= \int_0^\infty z\, 2\frac{\beta}{\alpha}\frac{z}{(1+(\beta/\alpha)z^2)^2}\,\mathrm{d}z \\ &= 2\sqrt{\frac{\alpha}{\beta}}\int_0^\infty \frac{y^2}{(1+y^2)^2}\,\mathrm{d}y = 2\sqrt{\frac{\alpha}{\beta}}\frac{\pi}{4} = \frac{\pi}{2}\sqrt{\frac{\alpha}{\beta}}. \end{aligned}$$

Problem 4

See Problem 4 of the Software Engineering examination, February 2024.

Transcription notes

1. The remainder after the second integration by parts should have a plus sign. The minus sign in the original conflicts with its next inequality.

2. After taking logarithms, the right-hand side should be $\ln(1/100)$ rather than 1%. The original's following line uses the logarithm.

3. The original repeatedly writes $\alpha\sqrt{n}$ for the standard deviation of the sample mean. For $X_i \sim \mathrm{Exp}(1/\alpha)$ it is $\alpha/\sqrt{n}$. The corresponding approximate bound is $c \approx 2.57\alpha/\sqrt{n}$.